KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE



PROJECT WORK TOPIC

FIRE PREVENTION SYSTEM FOR APARTMENT (AN IOT SIMULATION).

PROJECT OWNER INDEX NUMBER

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SEPTEMBER, 2026

DECLARATION

I hereby declare that this submission is the work of Nana Agyare Dennis for the final project of his two BSc academic years, and that, to the best of my knowledge, does not hold any article previously published or presented for certification, except for those which have been properly acknowledged and cited in this document.

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6505724		

DIDECATION

I dedicate this project to myself, for the determination, perseverance, and commitment demonstrated throughout the course of my academic journey. This work is a reflection of the effort, challenges, and growth experienced along the way.

I also dedicate this project to my family, whose love, prayers, encouragement, and unwavering support have been a constant source of strength and motivation.

To my friends, I express my appreciation for their encouragement, support, and companionship throughout this journey.

Finally, I dedicate this work to my supervisor for the guidance, patience, constructive feedback, and valuable contributions that have helped me successfully complete this project.

CERTIFICATION

As part of the Kwame Nkrumah University of Science and Technology award for a Bachelor of Science Degree in Information Technology, this project was supervised by Kate Takyi, Ph.D.

SUPERVISOR

KATE TAKYI, Ph.D.

SIGNATURE……... DATE……

ACKNOWLEDGEMENT

I thank God Almighty for his grace and mercies for my life and for being with me.

throughout the year. My sincere appreciation goes to my supervisor, Kate Takyi, Ph.D., for her outstanding support and knowledge and patience. I cannot forget my dear parents for their prayer, love, and hard work throughout my academic years and

Life as a whole. I also extend my sincere appreciation and gratitude to my friends for the academic support they offered to me throughout my academic years.

ABSTRACT

Nowadays, the term ‘’IoT’’ (internet of things) has become extremely important in our lives. This technology is used in many fields such as education, health, industries, agriculture, and infrastructures. To learn and understand how this technology works, many practical learning tools are used. The thesis aims to introduce IoT, a simulation tool where students can simulate or build and manage the systems for a better understanding of the philosophy behind networks. The tool used is Cisco Packet Tracer, which is software developed by Cisco that is used to create and simulate a virtual network, basically a wireless network, without the need for any network hardware. The tool is free of charge and suitable to work with almost all operating systems. Cisco Packet Tracer allows users to have practical networking technology knowledge. In this thesis, a "Cisco packet tracer" is used to design a fire prevention system for an apartment.

KEYWORDS: Cisco Packet Tracer, fire, prevention, apartment, IoT.

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ABBREVIATIONS

3GPP	3rd Generation Partnership Project
API	Application Programming Interface
APN	Access Point Name
ARPANET	Advanced Research Project Agency Network
CLI	Command Line Interface
DHCP	Dynamic Host Configuration Protocol
DNS	Domain Name System
GUI.	Graphical User Interface
IaaS	Infrastructure. as a Service
IoE	Internet of Everything
IoT	Internet of Things
ISP	Internet Service Provider
LAN	Local Area Network
NetAcad	Cisco Networking Academy
RFID	Radio Frequency Identification
RIP	Routing Information Protocol
SaaS	Software as a Service
POP3	Post Office Protocol version 3
Telnet	Telecommunications Network
SSH	Secure Socket Shell
DSL	Digital Subscriber Line
FTP	File Transfer Protocol

SMTP	Simple Mail Transfer Protocol
HTTP	Hypertext Transfer Protocol
TFTP	Trivial File Transfer Protocol
AAA	Authentication, Authorization and, Accounting
NTP	Network Time Protocol
SNMP	Simple Network Management Protocol
VOIP	Voice-Over-IP
SCCP	Signalling Connection Control Part
ISR	International Standard Recording
TCP	Transmission Control Protocol
UDP	User Datagram Protocol
BGP	Border Gateway Protocol (TCP/IP)
ICMP	Internet Control Message Protocol
IP	Internet Protocol
ARP	Address Resolution Protocol
NAT	Network address translation
GRE	Generic Routing Encapsulation
IPSec	IP Security
VPN	Virtual Private Network
HDLC	High-level Data Link Control
PPP	Point to Point Protocol
STEP	Spanning-Tree Protocol
PPPoE	PPP Over Ethernet

DTP	Data Transfer Process
VTP	VLAN Trunking Protocol
QoS	Quality Of Service
CDP	CRL Distribution Point [Microsoft
WEP	Wired Equivalent Privacy
SLARP	Serial Line Address Resolution Protocol
WAN	Wide-Area Network
WAP	Wireless Access Protocol
TCP/IP	Transmission Control Protocol/Internet Protocol
EAP	Extensible Authentication Protocol [Microsoft]
PKT	Packet Tracer file extension
PSK	Phase Shift Keying
SSID	Service set identifier

CHAPTER ONE

1.0 INTRODUCTION

The terms "internet of things" and "web of everything" allude to the idea that there may be things (objects) that may be meaningfully located, addressed, and detected using data-detecting devices (sensors) and controlled over the internet. Things are real objects with cutting-edge IDs that can transmit data via a network. Examples of genuine items include autos, cutting-edge mobile phones, home appliances, games, cameras, medical devices, contemporary frameworks, creatures, people, and constructions, among others (Thera, D. 2020).

The Internet of Things is another progressive and cutting-edge innovation where any article becomes a brilliant item and, where possible, imparts data without reference to human mediation. The Internet of Things is expected to significantly improve our lives by enabling us to carry out our commitments and obligations in a better way.

1.1 BACKGROUND OF THE STUDY

Many apartments in Ghana house tenants with varying years of stay, with one or two years being their minimum length. These tenants use electronic devices in their various rooms, they use LPG gas for cooking, and they use a variety of heating gadgets. These devices can set the premises ablaze when occupants, through human errors, forget to switch off heating gadgets.

Additionally, bad and defective electrical fittings can also cause fires that could destroy lives and property if undetected on time. Even though most of these apartments follow fire service standards and have enough fire extinguishers to put out any fire, most of the time, firefighters are called in.

We are unable to salvage lives and properties because there are no smoke notification systems in place to quickly announce the onset of fire for quick remedy.

1.2 PROBLEM STATEMENT

Bad and defective electrical fittings can also create fires that could destroy lives and property if undetected on time. Even though most of these apartments adhere to fire service standards and have provided enough fire extinguishers to put out the fire in any event, most times, firefighters are unable to salvage lives and properties because there are no smoke notification systems in place to quickly announce the onset of fire for quick remedy.

1.3 OBJECTIVES OF THE STUDY

The overall goal of this project is to develop a system that will be able to identify the beginning of a fire and announce it through IoT.

1.3.1 SPECIFIC OBJECTIVES

The particular goals are

1. Create a scalable apartment smart network with Packet Tracer.

2. Configure smart devices so that they can communicate effectively.

3. To configure a fire prevention agent to report the onset of fire on time

4. To recommend smart network prototypes to apartment developers and owners.

1.4 RESEARCH QUESTIONS

1. Is there an available smart apartment network?

2. Are there smart devices configured to communicate?

3. Is there a fire prevention agent to report the onset of fire promptly?

4. Is there a recommendation for a smart prototype network for hostel developers and owners?

1.5 NOVELTY OF THE PROJECT

A proposed smart, scalable apartment network will be built from scratch on the packet tracer test bed to mimic a real-life apartment. Smart devices will be configured to communicate with each other with user control, where a smart smoke detector will be configured to detect the smoke at the onset of a fire and report it in real-time to apartment occupants for necessary action. This prototype network will subsequently be recommended to apartment developers.

1.6 SCOPE OF THE STUDY

This research is limited to the configurations of smart devices like smoke detectors, smartphones, smart cars, water sprinklers, smart doors and windows, home gateways, etc. found on the Packet Tracer simulator and the requirements of apartment fire safety.

1.7 ACADEMIC AND PRACTICAL RELEVANCE OF THE PROJECT

At the end of this project, the recommended prototype network, if adopted by apartment developers, will aid in the reduction of gross fire hazards in apartment through timely smoke detection and reporting for immediate remediation.

1.8 PROJECT ACTIVITY PLANNING

This project is divided into many chapters. The significance of this research, its purpose, objectives, and goals, as well as the research topics it examined, are highlighted in the current chapter, which serves as the study's initial introductory chapter. The second chapter is a review of the literature that compiles data from several different sources.

The secondary data gathered in the second chapter and the available reference material were both crucial in helping to formulate the research's conclusion. The significance of the case study research has been stressed in the third chapter, which discusses the research technique. The third chapter offers details on various data collection methods and which ones are best for these. The case study discussion and analysis are covered in the fourth chapter, while the final chapter includes the.

CHAPTER TWO

REVIEW OF RELATED SYSTEMS

2.0 Introduction

This chapter briefly introduces concepts like IoT definition, history, applications, etc., and their reviews.

2.1 IoT explanation

The phrase "internet of things" or "internet of everything" alludes to the idea that every object may be able to be located, addressed, and controlled online. Things are genuine objects with fascinating identifiers that can transmit data via a network. Examples of genuine products include cars, cutting-edge phones, appliances, toys, cameras, medical equipment, contemporary construction, people, and other living things. Another innovative and forward-thinking technology is the Internet of Things, which allows any device to become intelligent and transmit data about itself without the need for human intervention. Our lives are expected to significantly improve thanks to the Internet of Things, which will help us fulfill our commitments and run errands.

2.2 History and Evolution of IoT

At Carnegie Mellon University in the 1970s, students studying the field of software engineering installed miniature switches inside Coca-Cola candy machines to test whether the machines' cooling systems were keeping the cans of soda cold enough and whether they were web-accessible. This was the very first instance of the Internet of Things. This first invention encourages additional research on networked machines. Intriguingly, John Romkey connected a toaster oven to the web with a TCP/IP convention later in the 1990s. (crsreports.congress.gov 2020). Scientists developed a framework to assess the amount of espresso that is available in the espresso machine in 1991 at Cambridge University. The idea was to use the webcam to snap three quick images of the coffee maker and send them to a local computer in their lab so that everyone could see how much coffee was being made. (crsreports.congress.gov 2020). The concept of the "web of things" (IoT) was first introduced by Kevin Ashton in a presentation paper written for Procter & Gamble in 1991. He explained the web of things as a technological advancement that links gadgets in the program. RFID innovation (radio recurrence and identifiable proof) is being used (crsreports.congress.gov 2020). At the first international conference on the internet of things in 2008, academics from 23 countries converged in Switzerland to study RFID, short-range remote communications, and sensor organizations. According to Cisco, the web of things truly began in 2010 because of the sheer number of linked devices. Furthermore, they claimed that the number of linked devices outnumbered the number of people on the planet (crsreports.congress.gov 2020). 6. The number of connected devices essentially surpassed the number of people in 2015. The graphic above shows that there will be 50 billion people on earth in the year 2020, making up 7.6 billion people in total. This demonstrates that there are far more related devices than there are actual devices. (crsreports.congress.gov 2020).



Figure 1.0. Cisco Prediction of Connected Devices

2.3 Applications for the Internet of Things

The breakthrough that will have the biggest impact on our lives is the internet of things. Numerous industries, including agriculture, energy-related health care, transportation, and others, use this concept. I'll illustrate some of the internet's things' usages in the current day and age in this section.

2.3.1 Internet of Things for Industry (IIoT)

The Internet of Things is used in the business world to improve efficiency and execution. This is known as the Industrial Internet of Things (IIoT). For example, Internet of Things devices can be used to monitor, regulate, and provide support for a processing plant; they can also be used to identify consumption inside a treatment facility pipe or predict equipment malfunctions so that maintenance can be provided before it is too late. The use of the internet of things in business will enable a variety of ventures, including production, food, auto, and other initiatives, to finish jobs without any issues.

2.3.2 Medical Internet of Things (IoMT)

The industry that will contribute most to the Internet of Things innovation is the medical one. Through the use of cutting-edge devices like heart monitors and speed creators, the internet of things in medical services enables specialists to manage patient circumstances from anywhere at any time across a network to provide checking, examination, and remote settings. Numerous more internet of things devices, such as smartwatches and wellness trackers, can be used to manage our health.

2.3.3 The Smart City

Smart cities refer to places where the Internet of Things devices are used to monitor and operate the city's systems for transportation. Internet of Things devices can also be used in smart cities to manage other aspects of daily life besides transportation and infrastructure, such as managing water quality, analyzing and monitoring the energy grid, and other activities.

2.3.4 The Smart Home

A "smart home" is a home that has smart appliances such as a refrigerator, air conditioner, light, camera, fan, smart interior regulators, entryway locks, and more that can be controlled and managed remotely via the web using a PDA or PC. The ability to manage household equipment remotely provides security, convenience, and ease to mortgage holders. Innovative home assistance saves energy and helps to prevent a few accidents.

The mortgage holder can remotely monitor the camera, home caution system, and location framework to see if there has been any infringement for security reasons.

2.3.5 The Smart Car

A "smart car" is a vehicle that uses multiple sensors to enable all of its functions to be remotely controlled by a computer or a high-tech smartphone. We may examine the vehicle's oil level and radiator water and, in any case, be able to operate the vehicle from a distance with the help of this particular web of things application. (Flifel 2019).

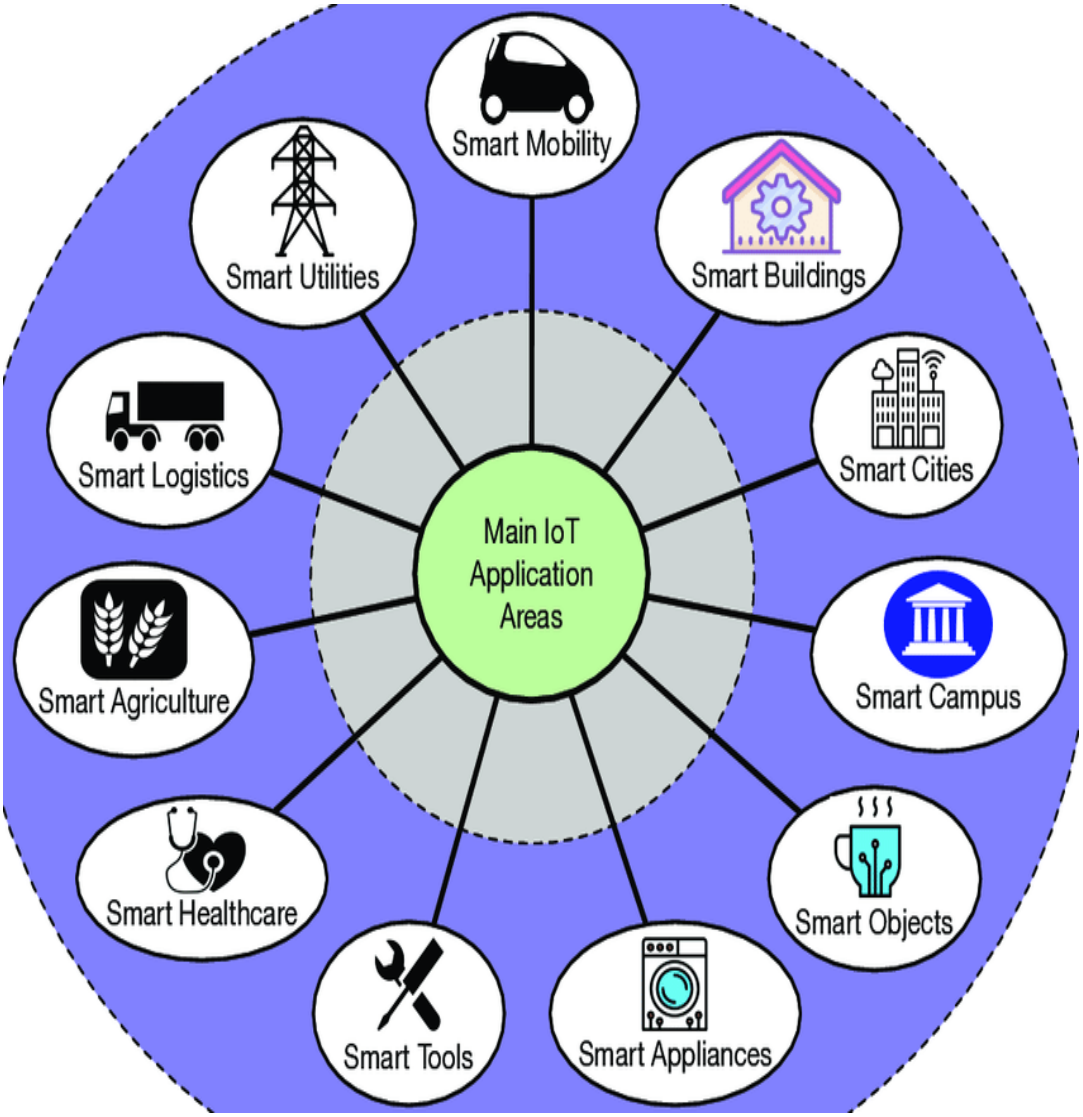


Figure 1.1 IoT Applications

2.4 Overview of Cisco Packet Tracer

The Cisco Packet Tracer is an effective virtual organization learning and understanding tool for PC organizations. Cisco developed the tool to make it possible for customers or students to obtain important information about systems management technology. Users or students can simulate and plan an organization using virtual devices like hubs, routers, switches, and so on. Without any organization, the simulation in Cisco Packet Tracer functions.



Figure 1.2 Cisco Packet Tracer Interface

2.5 Workspaces for packet tracers

There are two workspaces in the Cisco packet tracer, one of which is physical and the other logical. While the physical view provides a graphical representation of the virtual organization gadgets, the logical view allows clients to place and interface with virtual organization gadgets. From a physical standpoint, the devices have accessible openings where we may install more modules. This particular simulation device's advantage is that it offers a setting where gadgets resemble those in real life. This is crucial because it gives customers the chance to learn everything there is to know about devices before using the actual gear.

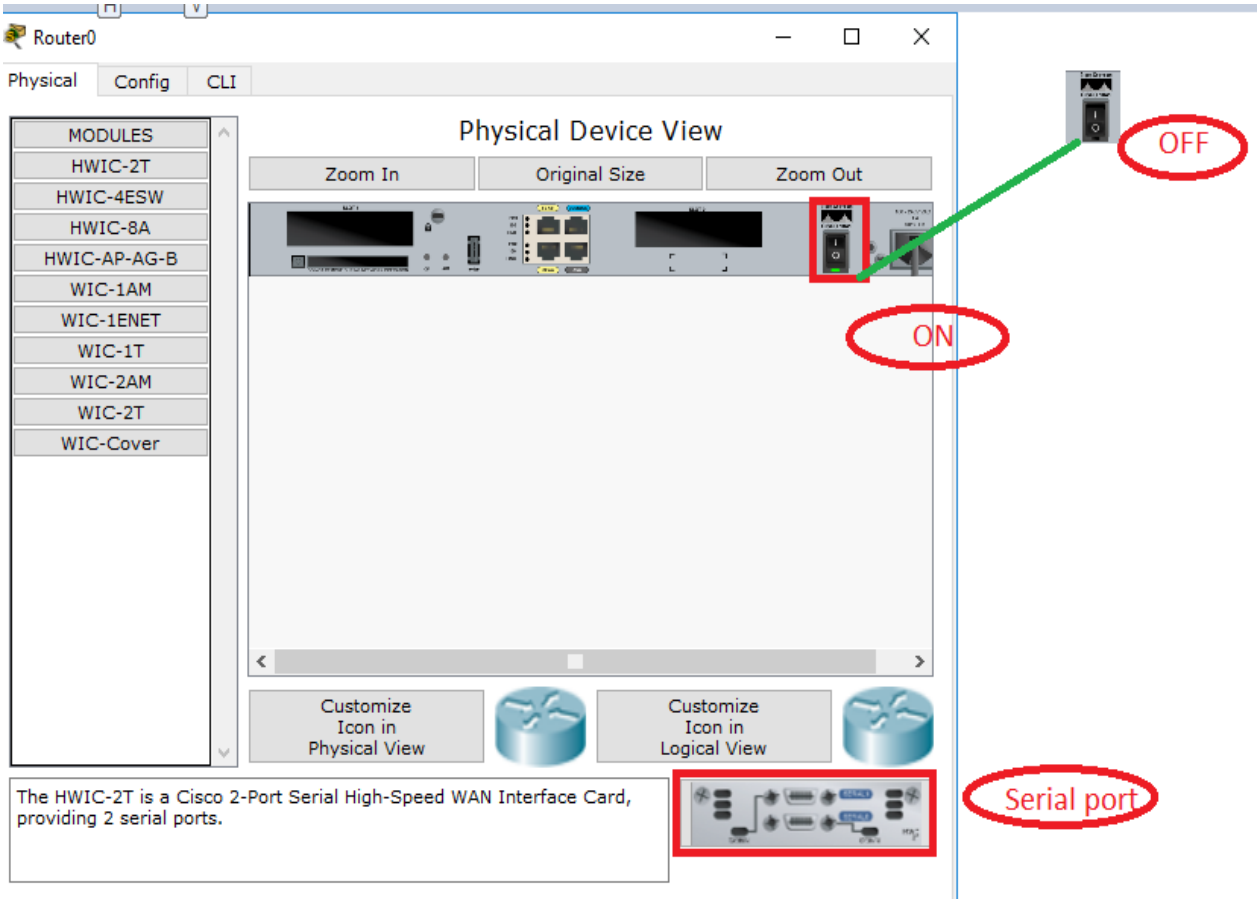


Figure 1.3 Physical View of ISP Router

2.6 Packet Tracer Mode

The device has two additional modes: constant mode and reenactment mode. Students or clients can have a clear understanding of how the devices operate consistently. In this mode, devices function as actual contraptions. On the other hand, the reproduction mode aids students or clients in understanding the central idea behind association tasks. This mode enables the customer to visualize the distribution of data across an organization and control periods.

2.7. Cisco configuration techniques

We have two options for setting up devices using the Cisco packet tracer: the config tab or the CLI tab (command line interface). Using the Cisco order line, we organize devices using the order line interface. Utilizing the order line connection point has the advantage that the orders we use to create gadgets are essentially the same orders we use for real gadgets. An example of a switch design using a command line interface is shown in the figure below. (CLI).

No Cisco order information was required for the configuration tab. A graphical point of interaction is used to complete the design using the configuration tab. When a client has more time and needs to quickly assemble devices, this design strategy might be used. This tactic can enable us to speed up the design process. (2019, Kumar, Krishna, and Ramprakash).

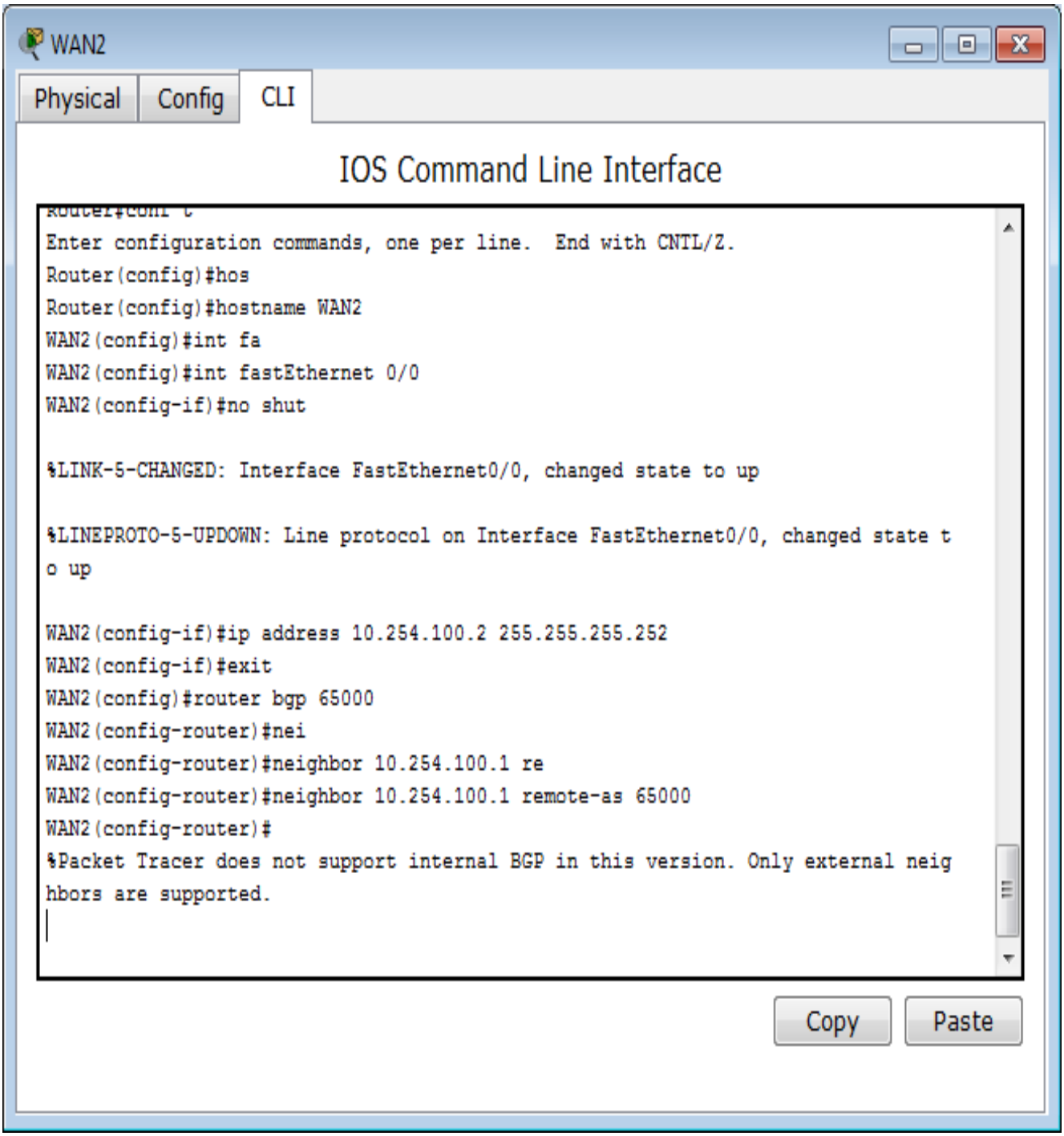


Figure 1.4 Cisco Packet Tracer Command Line Interface Tab

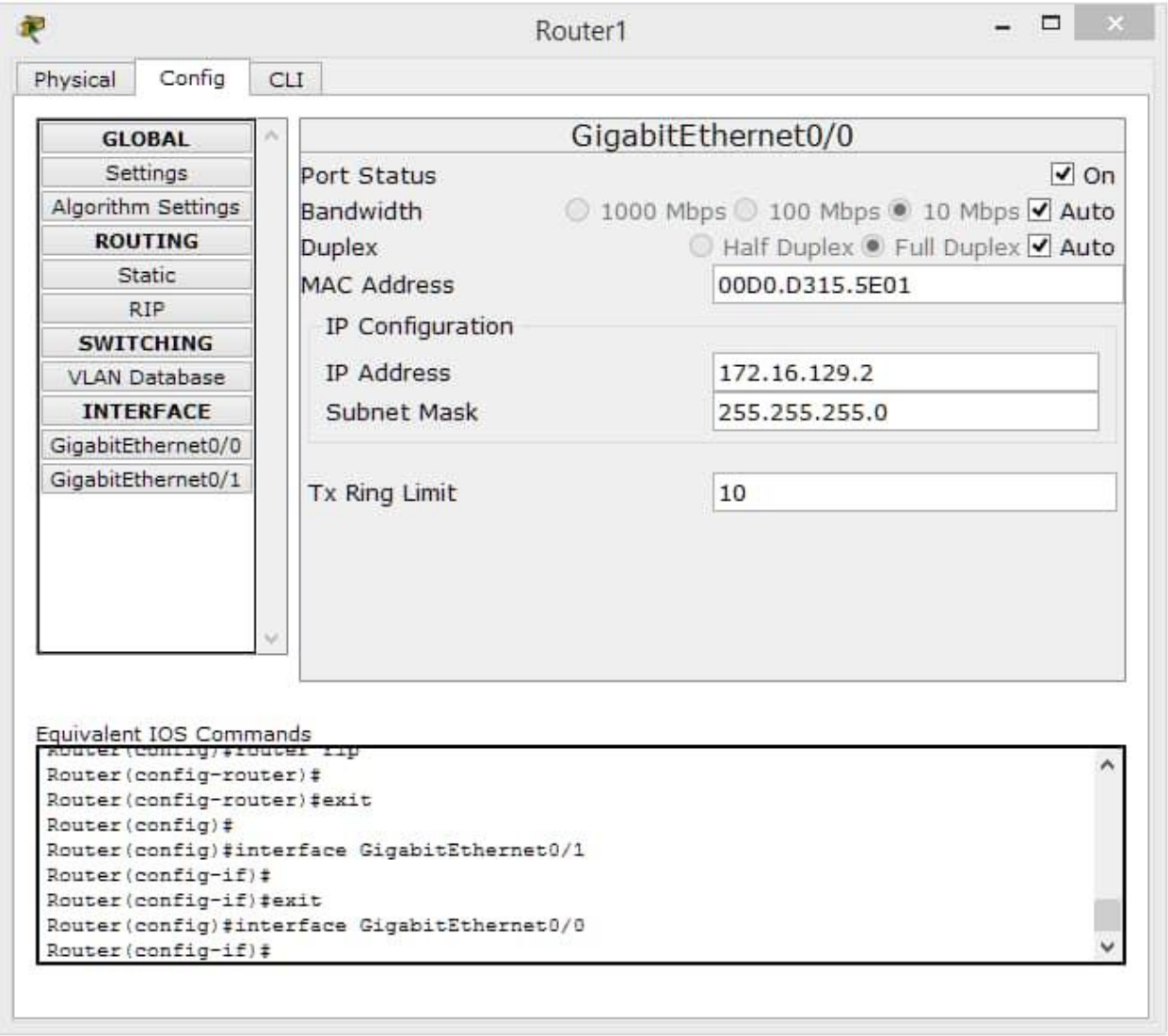


Figure 1.5 Cisco Packet Tracer Config Tab

2.8 protocols that Cisco Packet Tracer supported

Numerous conventions are supported by Cisco packet tracers. The table below displays the groupings of conventions that parcel tracers uphold. Patel 2016), (Kumar, Krishna, and Ramprakash, 2019).

Table 2 – Protocol Supported by Cisco Packet Tracer

Layer	Supported Protocols
Application	FTP, SMTP, Telnet, SSH, DNS, DHCP, VOIP.
Transport	TCP and UDP, RTP, IP Fragmentation.
Network	BGP, IPv4, ARP, CEF, IPSec, PTP, NAT.
Network Access/Interface	WPA, VLANs, DSL, EAP, VTP, CDP.

2.9 Internet of Things and Cisco packet tracer

The most recent version of Cisco Packet Tracer added a new feature that can help us with the web of things entertainment. Smart devices, such as sensors, actuators, and microcontrollers, are among the new elements. (Saha, Ishraque et al., 2019). Smart windows, smart fans, bright lights, and warning alarms are a few of the innovative devices associated with parcel tracking. We can also find a few sensors, such as those for carbon dioxide, temperature, stickiness, and water level. (Kumar and Krishna 2018), (Saha, Ishraque et al. 2019). The most major change in the new version is that all of the devices may now be programmed using blocky, Python, and other programming languages. Additionally, they can typically be connected through a wired link or remotely. The new packet tracer offers a variety of cabling options, including copper straight links, copper hybrid links, fiber-fast Ethernet links, and IoT bespoke links. In the future, we can also choose auto cabling, in which case the devices will choose the proper link to connect two devices. (2019, Kumar and Krishna). The web of things devices in the Cisco Packet Tracer can be used to build and simulate various web-of-things applications, including smart cities, smart industries, and smart homes. The benefit of using Cisco Bundle Tracer is that users can interact with the devices just like they would with real ones. Additionally, thanks to its multiuser functionality, many users can

work together to create a virtual organization by using an existing one. (2019, Kumar, Krishna, and Ramprakash). To implement a clever house- or web-based home robotization framework, this postulation study is merely focusing on utilizing the innovative devices in the new form of the parcel tracer. A few other devices from the latest Cisco Packet Tracer are on display.



Figure 1.7 Cisco Packet Tracer Smart Devices

3.0 Types of Simulation Software

I. Network Simulator Version 3 (NS3)

NS-3 is a discrete-occasion network test system that is primarily intended for use in research and instruction. Ns-3 is open-source software that is permitted under the GNU GPLv2 license and is unrestrictedly available for review, development, and usage. The NS-3 C++ library provides many business recreation models that are implemented as C++ objects and wrapped in Python. Clients often connect to this library by creating a C++ or Python application that launches many replication models to create the important reenactment situation, enters the reenactment primary circle, and exits after the reenactment is complete. (Siraj, S., Gupta, A., & Badgujar, R. 2012)

Pros:

• The framework has been modularized.

• To consider modular libraries

• Individual modules contain catalog structure.

• To permit the node to utilize outside steering

Cons:

• Ns3 suffers from a lack of validity.

• Modules based in part on NS2

• Ns3 requires many maintainers.

Dynamic maintainers are required.

II. GNS3:

An enormous number of business engineers use GNS3 to simulate, build, test, and research real and virtual businesses. GNS3 enables you to run small geography made up of many devices on your computer, in contrast to others who have many devices hosted on several servers or perhaps on the cloud.

Pros:

• GNS3 is a free and open-source program.

• Capable of simulating complex organizations

• Re-enacted geography can be linked to the real world.

Cons:

• Slow throughput in comparison to real-world equipment;

•extremely limited interchangeability

•Excessive CPU usage

III. Packet Tracer from Cisco:

A popular network simulation tool used by Cisco Systems is Cisco Packet Tracer. To create, plan, build, and test your organizational problems in a wholly virtual biological system, you can create a simple or complicated network inside of Packet Tracer. Users of this network simulation tool can create network topographies and replicate them in existing PC organizations. One of the most popular network simulation programs among aspiring and beginning systems administrators is Packet Tracer.

Pros:

• Easy to set up

•Maintains Cisco switch, switch, and PC simulations

•Suitable for CCNA studies

•Acts out various gadgets and conventions (switches, for example).

Switches, remotes, RADIUS, etc.

•Free (requires enrollment at Cisco's NetAcad site)

Cons:

• Exclusive code—not open source

•Only reenacts Cisco gadgets (not valid)

Cisco pictures)

• There is no multivendor support.

• Incompatibility with genuine devices

3.1 Conclusion

The goal of this project was to use a Cisco packet tracer to simulate the internet of things. Another recent and cutting-edge development is the Internet of Things. Therefore, it was essential to create a virtual beneficial tool where students could learn about and comprehend this advancement. That is the inspiration behind this study. We chose the Cisco packet tracer because it provides a simulation environment for devices that look like devices in reality but aren't. In addition, the new version of the packet tracer includes various Internet of Things devices, actuators, and various sensors, making it the ideal test system for the internet of things.

The idea was to use Cisco Packet Tracer to implement and replicate a highly important Internet of Things application, fire prevention framework.

CHAPTER THREE

METHODOLOGY

4.0 INTRODUCTION

This chapter seeks to describe the simulation strategy, analyze the system, architectural design, and the empirical techniques applied in the development of the proposed project.

4.1 SYSTEM ANALYSIS

The most typical method of obtaining information, identifying problems, and breaking down a system into its component elements are needed for the system analysis step. System analysis is intended to focus on a system or a component of it to identify its objectives. The significant steps engaged in this stage incorporate characterization of the client's prerequisites and concentrating on the current system to verify the problem.

4.1.1 Requirement Analysis

Requirement analysis is the method involved in analyzing the expectations of the clients for an application or system that will be built or changed. It requires all the tasks that are conducted to distinguish between the needs of various stakeholders. There are three (3) kinds of requirements, to be specific: business requirements, user requirements, and system requirements.

4.1.2 Characterization of Current System

The ongoing system has the following attributes or elements:

- Lacks a central server or data sets to keep track of devices.
- Gadgets need the presence of people to control them.

- No security or authentication components are set up.
- Devices can’t be accessed and controlled from a distant area.
- has no internet availability.

4.1.3 Requirement Gathering

Requirement gathering looks to figure out what the proposed project needs to accomplish and what should be done to get that going.

4.1.4 Functional Requirements

Functional requirements look to determine the help the proposed system will give the client in satisfying their work tasks. Functional requirements specify what the system must do. The requirements listed below are the functional requirements of the proposed system.

- Allow tenants and apartment owners to access, monitor, and control devices from afar using a smartphone.
- Validate and authorize apartment owners and tenants to use smart devices.
- The system will detect smoke to prevent fire.
- The system will allow apartment owners to edit and remove faulty devices.

4.1.5 Non-Functional Requirements

The features or properties the proposed system must possess are known as non-functional requirements, or how it should be built. These are the properties that make the system attractive, fast, and reliable. Below are the non-functional requirements of the proposed system.

- ❖ The system will have a pleasing GUI.
- ❖ The system will score high on usability tests.
- ❖ The system will be user-friendly.

4.2 MODEL OF THE EXISTING SYSTEM

The existing systems are made up of devices found in the hostels that are manually managed and controlled by apartment owners or students. Since the systems require the presence of individuals to operate on them, it was very difficult for devices to be accessed, monitored, and controlled remotely over the internet. Apartment devices are not registered to a central server or connected. Devices were operating electrically independently. Security mechanisms were absent. This made it difficult to detect the onset of fire. There were no sensors or detectors to detect environmental conditions such as temperature. The primary way of preventing fire outbreaks from occurring was by using fire extinguishers.

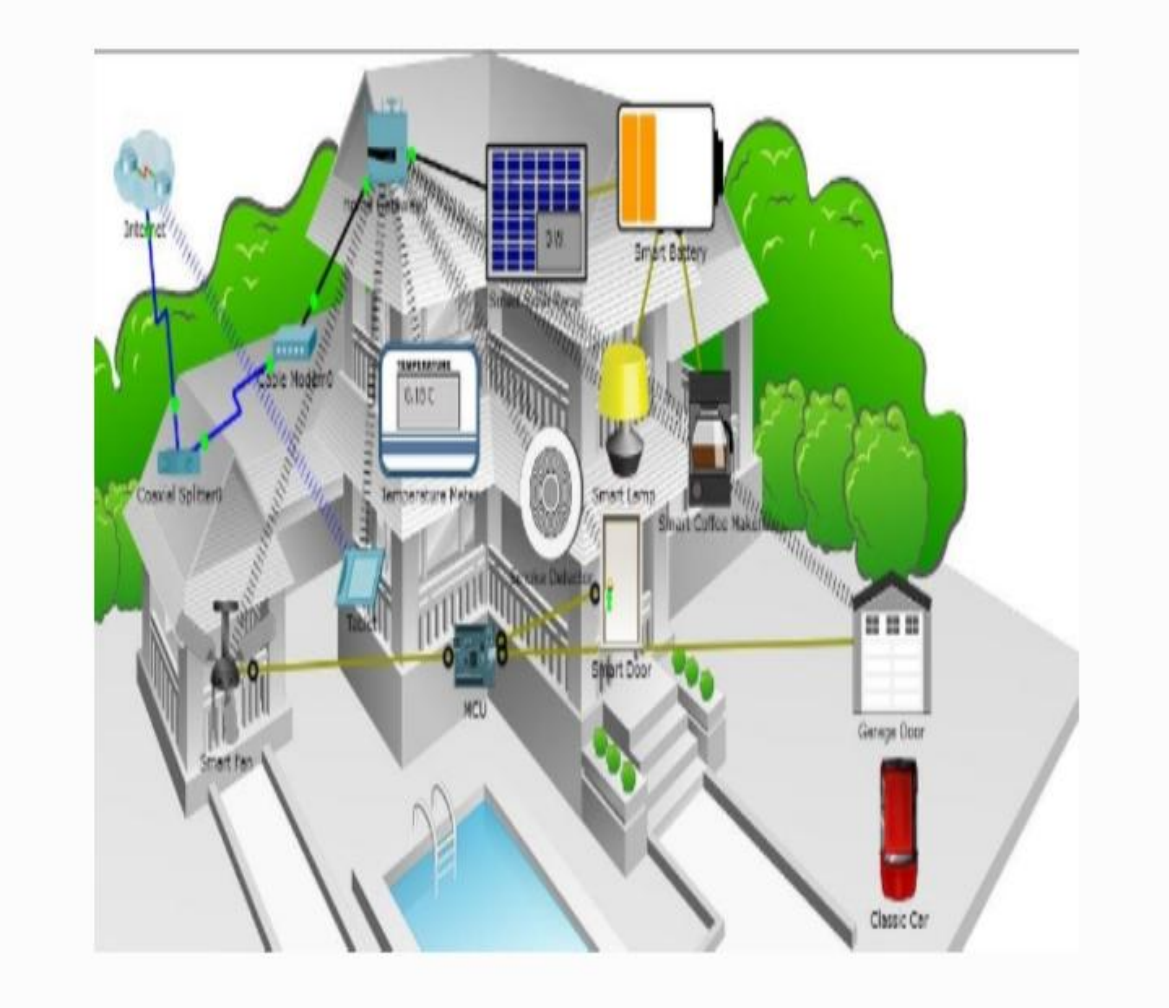


Figure 2.0 model of an existing system

4.3 SYSTEM DESIGN

The process of establishing a system's components, including modules, architecture, interfaces, and data, based on predetermined requirements is known as system design. System design refers to the definition, development, and design of a system that satisfies the unique needs and requirements of a user, a company, or another entity. Designs can be expressed using textual or graphic modeling languages. The design processes involve physical, logical, and architectural design.

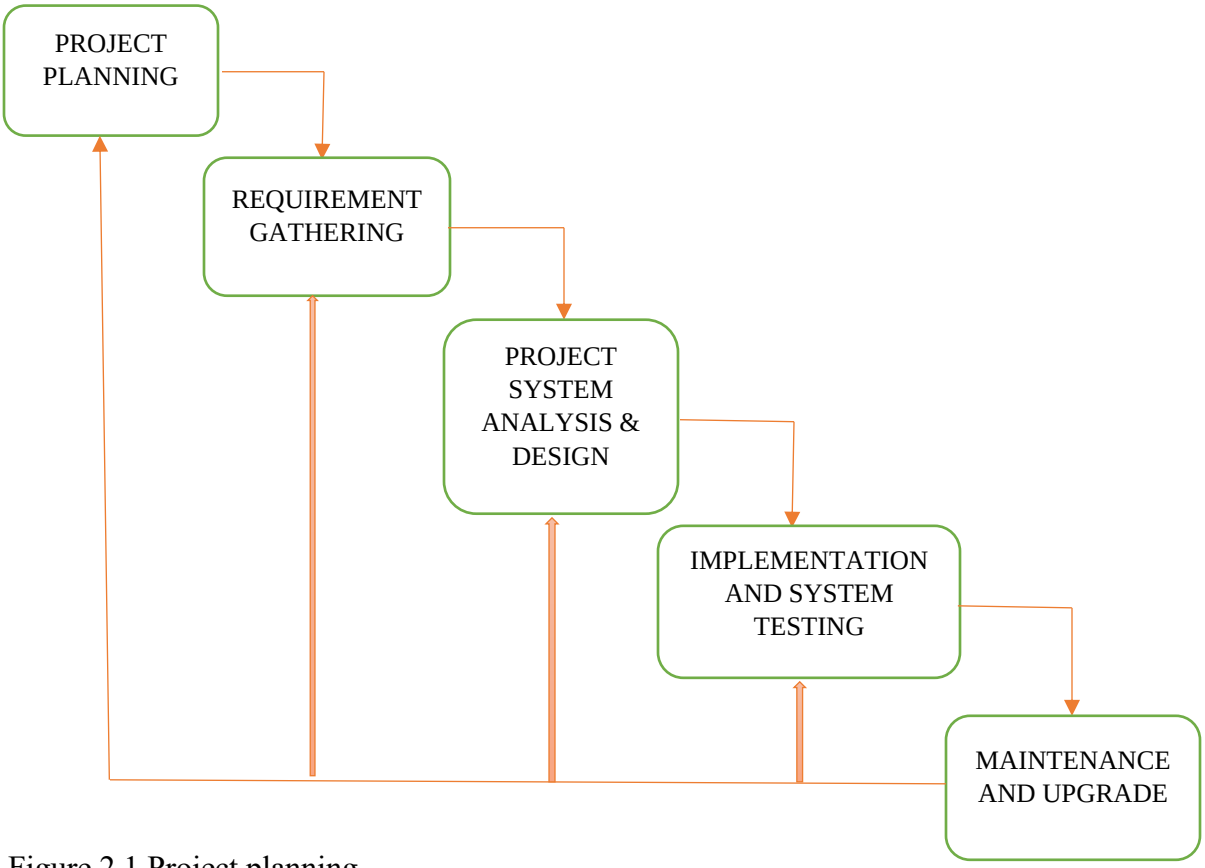


Figure 2.1 Project planning

The figure above is accumulating on project planning, which necessitates project planning, project system analysis and design, implementation and system testing, and system maintenance. Five different phases need to be improved, and there is a system development life cycle to illustrate how. The apartment, as well as the smoke detection and fire prevention system, saves the proposed system users' information so that they can understand how the project will be planned as well as the project's goals and objectives. Following that, substantial research on how to construct a good system is required, which can be established through a systematic review. The system was then created for the initial sample and fine-tuned to support their recommendations. Until the analysis, design, and implementation phases were iterative, users and designers agreed on the final details of the system. At this time, the project's final implementation phase can begin.

4.4 SYSTEM FLOWCHART

The flowchart of the system displaying the steps of a process or algorithm, different box types, and their arrows connected by this sequence can be a sort of picture. Such a diagrammatic display cannot gradually solve the issue. This process is illustrated in the box, and the linked arrows reflect the control flow. In several domains of design, flowcharts are familiar with the management of a process or program. Many input and output symbols are used, and the flowchart is representative.

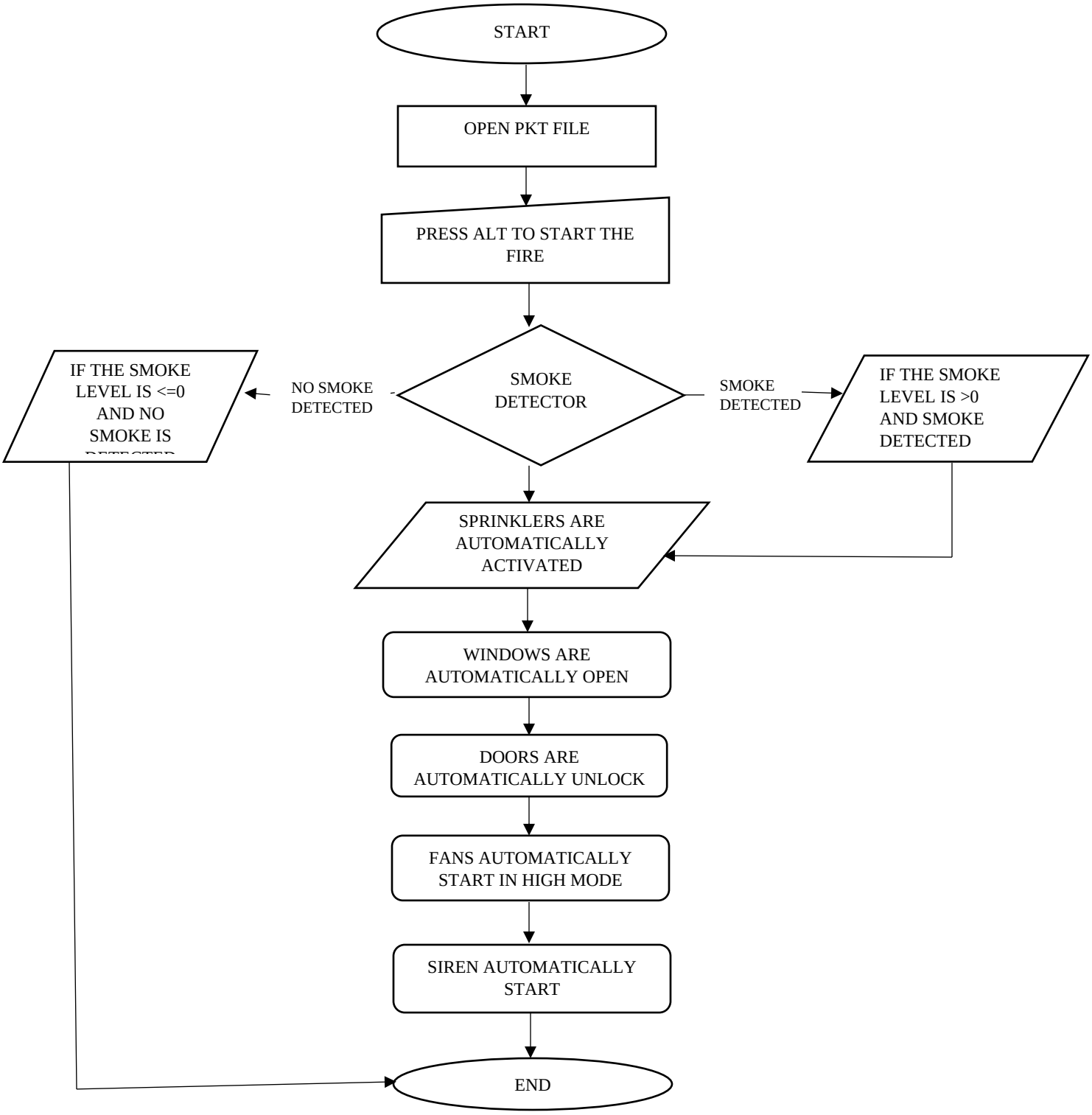


Figure 2.2 Fire Prevention System Flow Chart

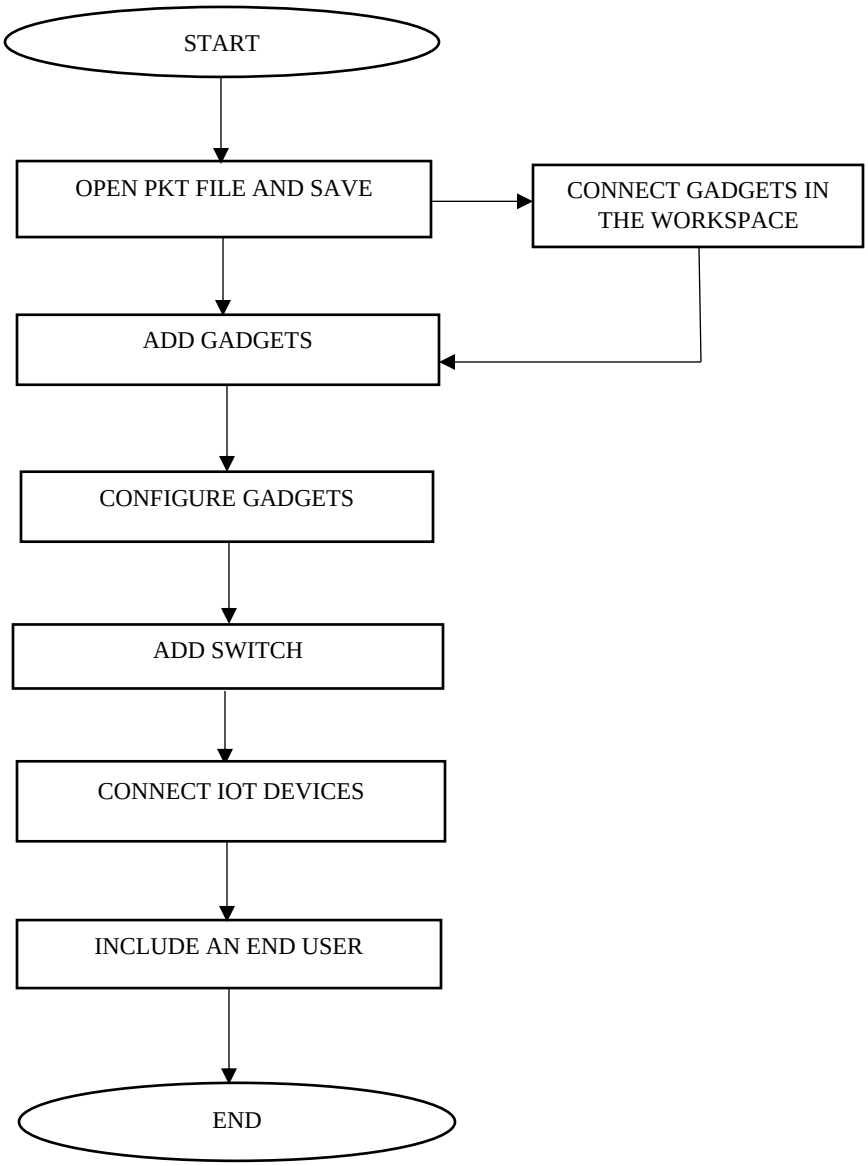


Figure 2.3 Device Configuration Flow Chart

4.5 Model of the Proposed System

The proposed smoke detection and fire prevention system was designed using the new release of Cisco packet tracer simulation software. The proposed fire prevention system was made up of smart IoT devices and networking devices connected using a cable medium (thus, wired or wireless) and configured with internet protocol addresses to aid in device communication. IoT devices are remotely managed and controlled by apartment owners or tenants from a distance. Since the system does not necessarily require the presence of individuals to operate on them, it is very easy and efficient for home devices to be accessed, monitored, and controlled remotely over the internet at any time. Home devices are registered to a central home gateway, which allows individuals to view the status of each appliance, whether it is active or inactive. The presence of sensors or detectors to detect environmental conditions such as temperature and give a response to users. The proposed fire prevention system for the apartment seeks to enable the remote accessibility of IoT devices such as ceiling fans, lamps, etc., improving security, safety, and comfort on the part of individuals and enhancing energy utilization within the apartment.

The diagram below represents the model of the proposed system using Cisco packet tracers.

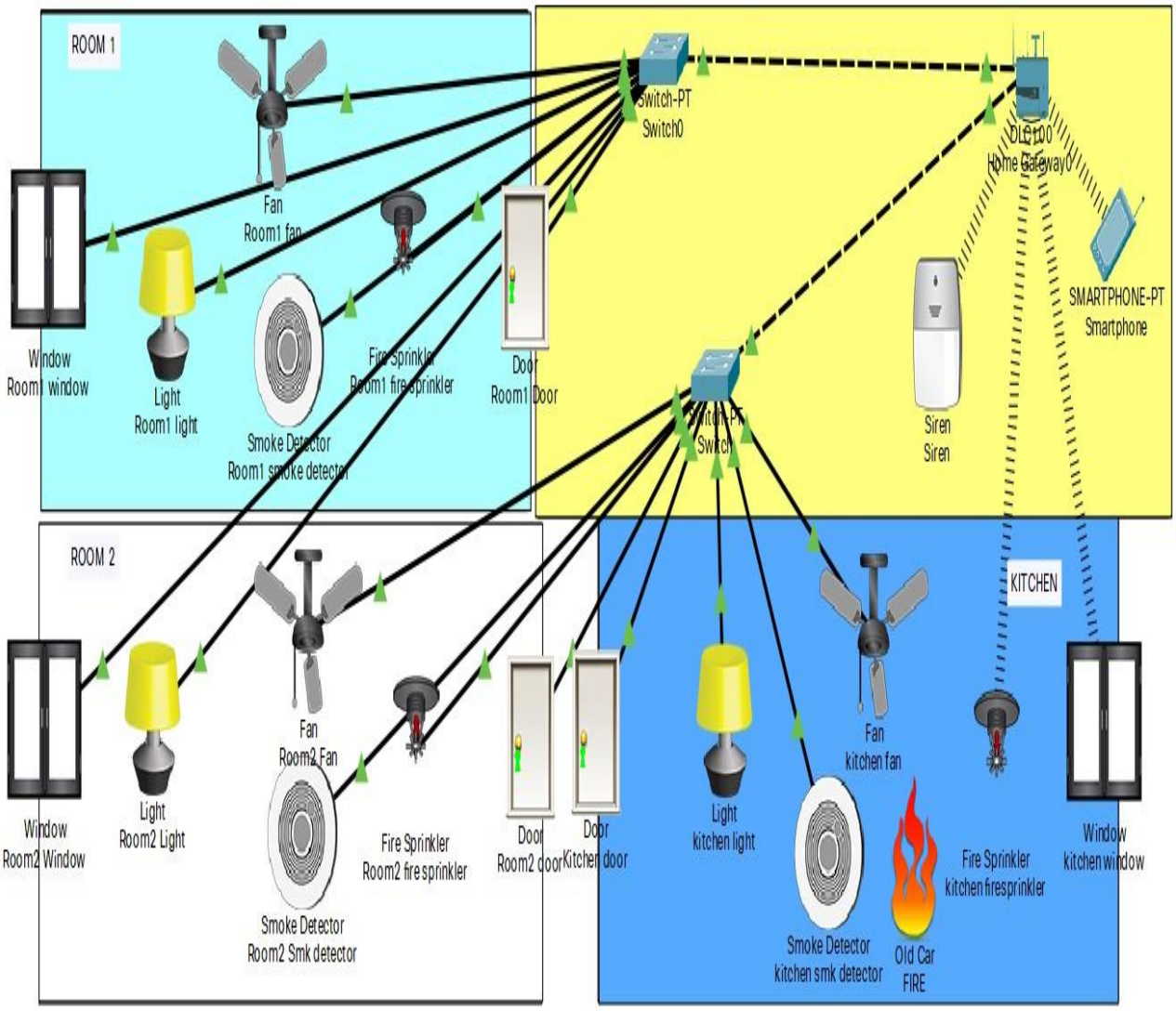


Figure 2.4 Model of the Proposed System

- The above-mentioned fire prevention system for the apartment is made up of four (4) components: a home network, the internet, and a 3G/4G network.
- Home Network: The home network comprises various Internet of Things (IoT) devices such as smart fans, RFID doors and windows, smoke detectors, and a home gateway. The Home Gateway serves as a connection point and a registration server for the various IoT devices.
- Internet: To connect to the internet, a link is made possible via the internet between the Internet Service Provider (ISP) Router and the Home Gateway via cable modem.
- 3G/4G Network: To remotely monitor, access, and operate the IoT devices connected to the home network, a smartphone must be connected to the closest cell tower.

4.6 PROTOTYPE

A prototype is simply a first version, model, or item created to test a concept or cycle. Rather than using a hypothetical framework, prototyping provides details for a real-life, functional one. The suggested smart home system was created using the Cisco packet tracing simulation software, as was previously stated. As a result, the prototype that was used had poor fidelity. It is an efficient, straightforward method of turning an idea into a more accurate depiction of what the finished item would look like. As compared to high-fidelity prototypes, which cost a lot since the actual physical devices are not being used, a low-fidelity prototype is characterized by its lower cost and its time efficiency since the devices used in the development of the proposed system are already found in the Cisco packet tracer.

The figure below depicts the prototype of the proposed system.

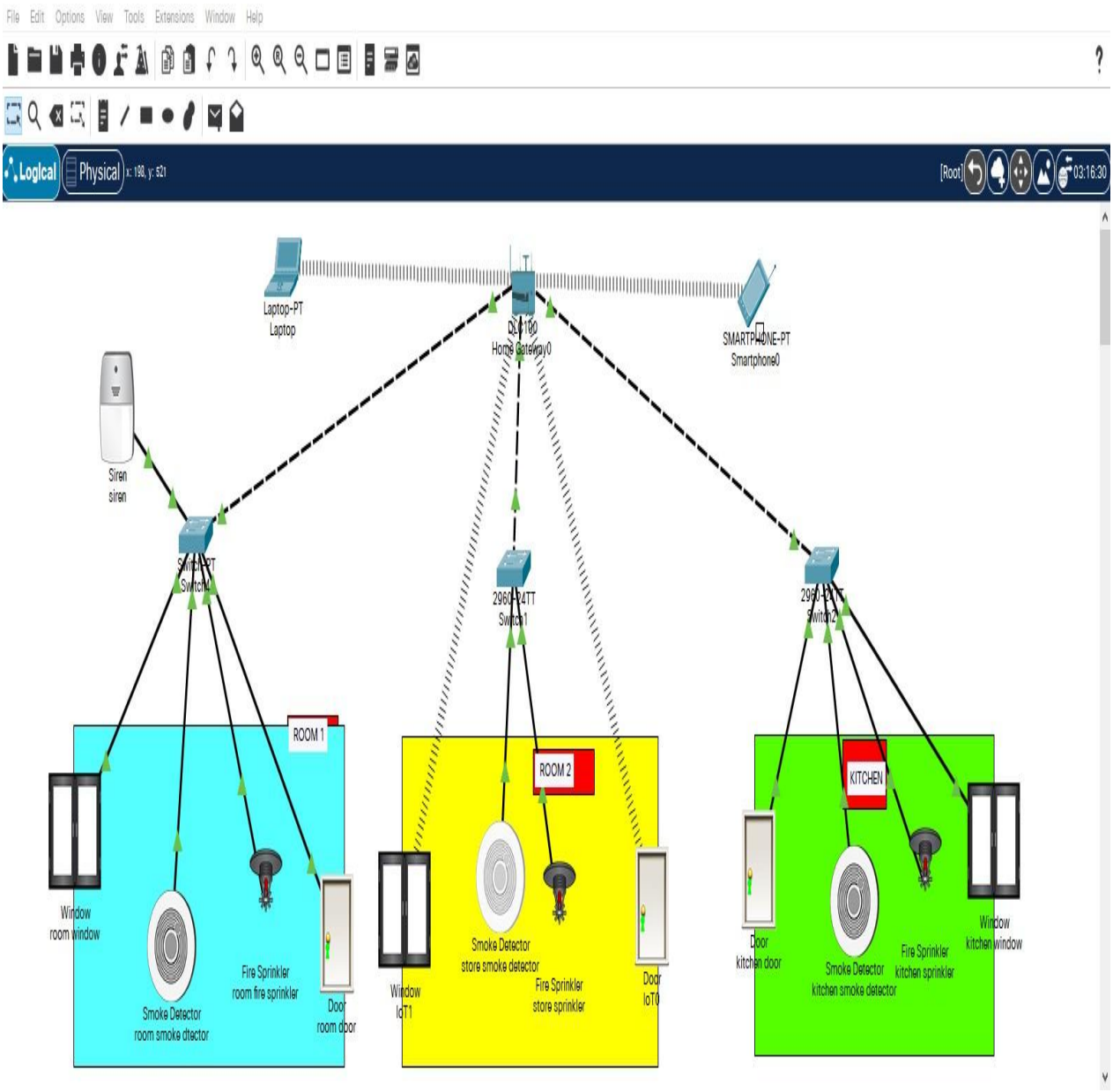


Figure 2.5 A Prototype of the Proposed System

4.7 INTERFACE

The front-end application view, known as an interface, is what clients and associates use to interact with the product. Through the user interface, clients can control both the product and the machinery. Today, UI can be found pretty much everywhere there is advanced innovation, including computers, cell phones, cars, music players, airplanes, ships, and so on. The UI is crucial for programming and is designed in a way that makes it natural for the client to grasp the software. The UI sets the stage for human-PC interaction. UIs can be graphical, text-based, auditory, or video-based depending on the hidden hardware and software configuration. A UI might be software, hardware, or a combination of both. The interface of the system is shown in the figure below. (CLI, C. L. I., 1997).

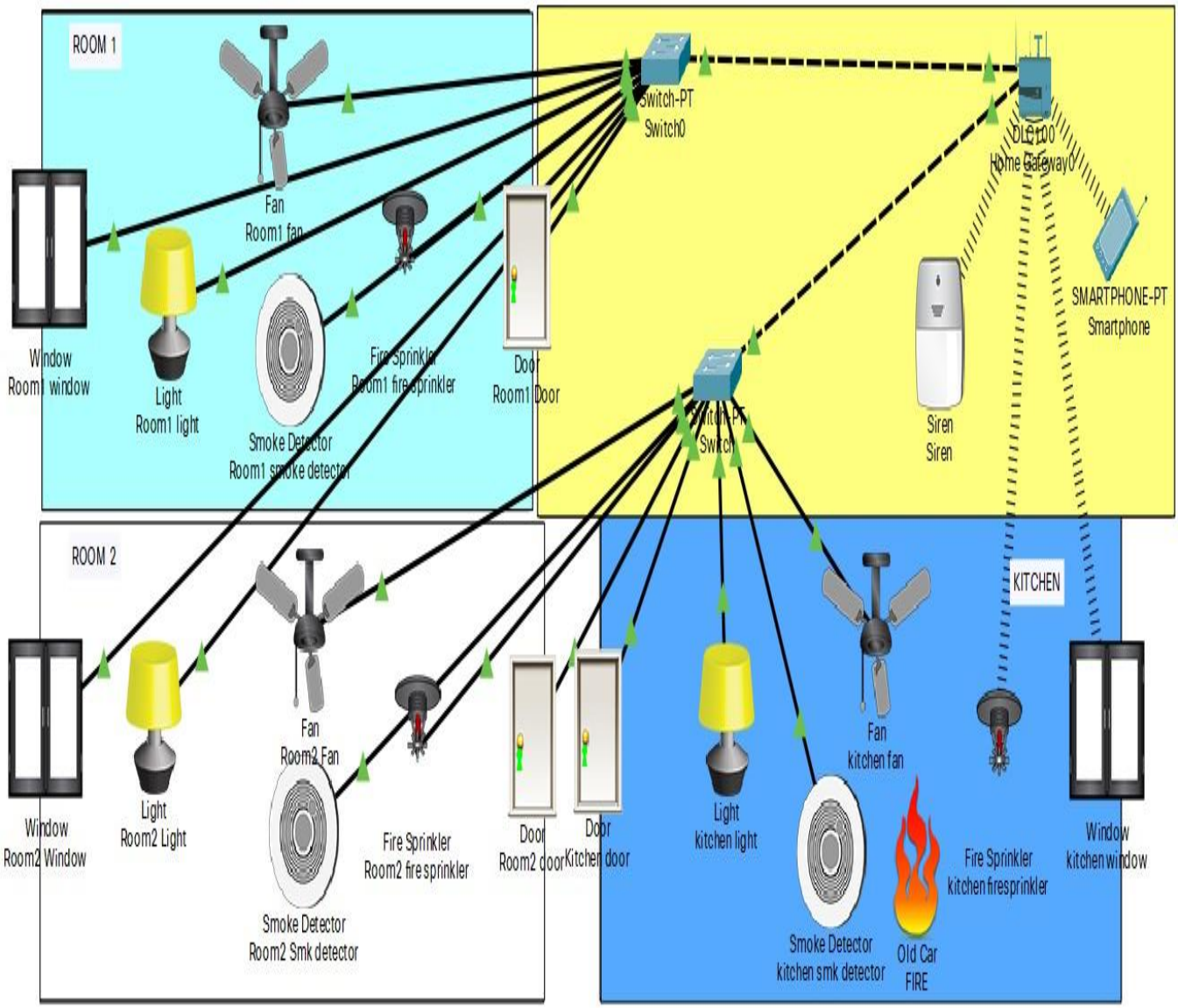


Figure 2.6: Interface of the developed system

4.8 CONCLUSION

This chapter of the project spoke about the existing systems and how, having analyzed the previous or existing ones, this project can bring out a system best suited to eradicate the emerging fire outbreaks in society. The chapter reviewed models of the existing systems and made mention of our system, which will be developed soon.

CHAPTER FOUR

IMPLEMENTATION, TESTING AND RESULTS

5.0 INTRODUCTION

This chapter briefly explains the overview of the Fire Prevention System for Apartments. It talks about the various methodologies used in developing the system using the Cisco packet tracer.

5.1 Fire Prevention System for Apartment Overview

An apartment is furnished with various smart items, for example, smart fans, smart lights, brilliant windows, and so on, that can be somewhat controlled utilizing a cell phone or PC through a web association. The framework offers the apartment occupiers convenience, reserve funds, security, and solace. Saving because the utilization of a few smart items, for example, smart lighting, can aid in energy conservation (reduce energy consumption) and cost reduction. Security is one of the biggest benefits of automation because all tasks are completed automatically. smoke detection and fire prevention system, since you can remotely control the gadgets and check whether there is a risk wherever you are in your home. Because of the opportunity it provides, consider checking to see if your lights are turned on and turning on your fan to cool yourself before going to your house. The apartment fire prevention system permits apartment occupants to deal with all the apartment gadgets anyplace whenever (Chattoraj, S. 2015).

5.2 Fire Prevention System for Apartment Implementation

The implementation of the fire prevention is done with the new version of the packet tracer, precisely Packet Tracer 8.0.1. Table 1.1 below shows the lists of the different devices used for the simulation and their functions.

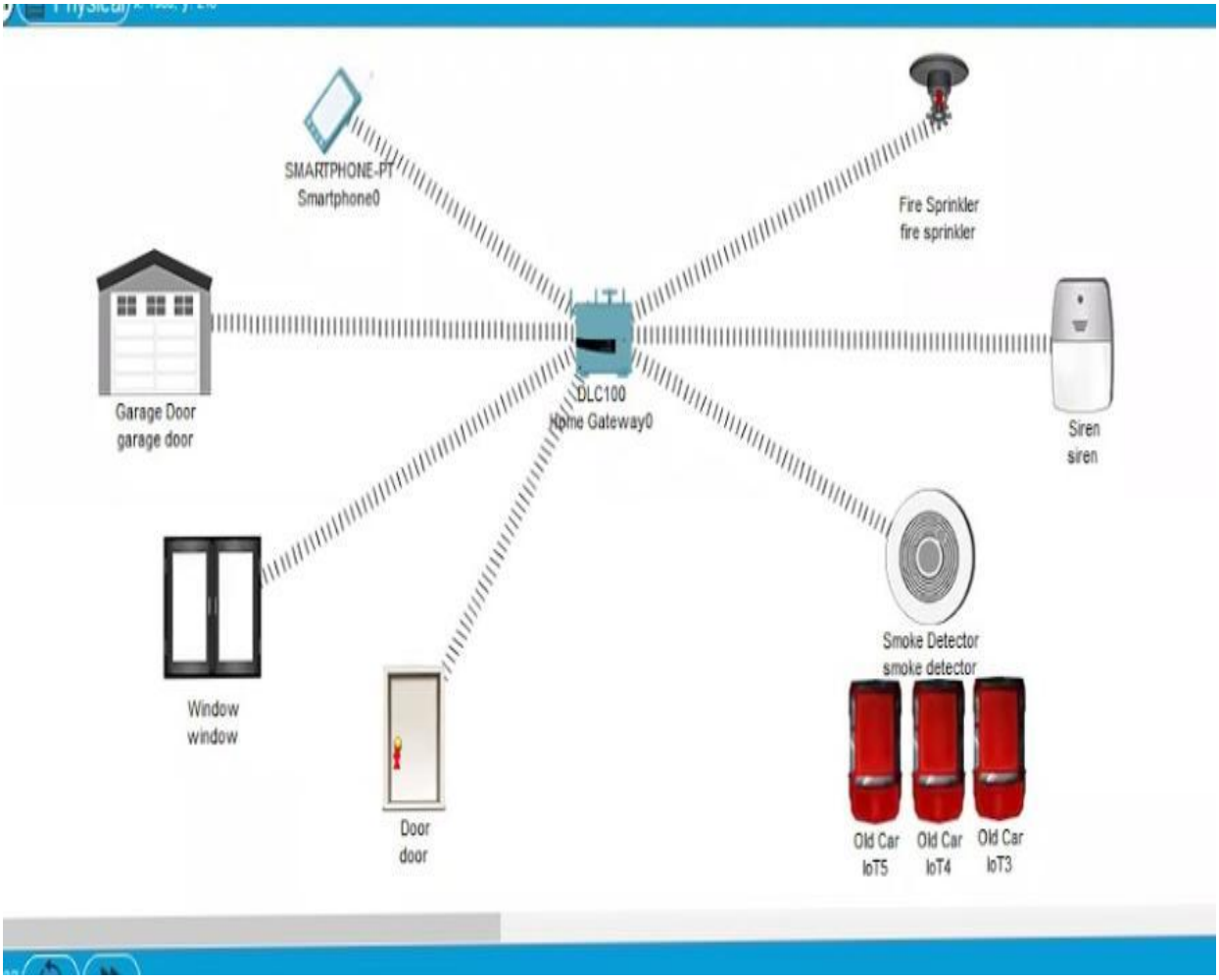


Figure 3.0 Prevention System for Apartment Application

Table 3 - Devices Used for the Simulation

No.	Devices	Functions
1	Home gateway (Delci 100)	Used to register smart gadgets
2	LED lighting	Utilized to illuminate the hostel
3	Smart door	Used to remotely open and close doors
4	Smart windows	Used to operate the windows remotely
5	Smart siren	Used to alert hostel occupants of emergencies
6	Fire sprinkler	Utilized to sprinkle fire
7	Fire	Used to start the fire
8	Switches	Used to concentrate on smart devices
9	Smartphone	Used to control smart devices from a distance
10	Smoke detector	Used to detect smoke at the onset of fire
11	Smart Fan	Used for ventilation in the hostel
12	Siren	To audibly give a sound alert on detection of smoke

5.3 Device Configuration

The simulation of the fire prevention systems for apartments using Cisco packet tracers is shown in Figure 1.2 below and is based on the methodology provided on the previous page. The simulation includes the home door, switches, IoT (web of things) remote server, PDA, smoke alarm, and vehicle.

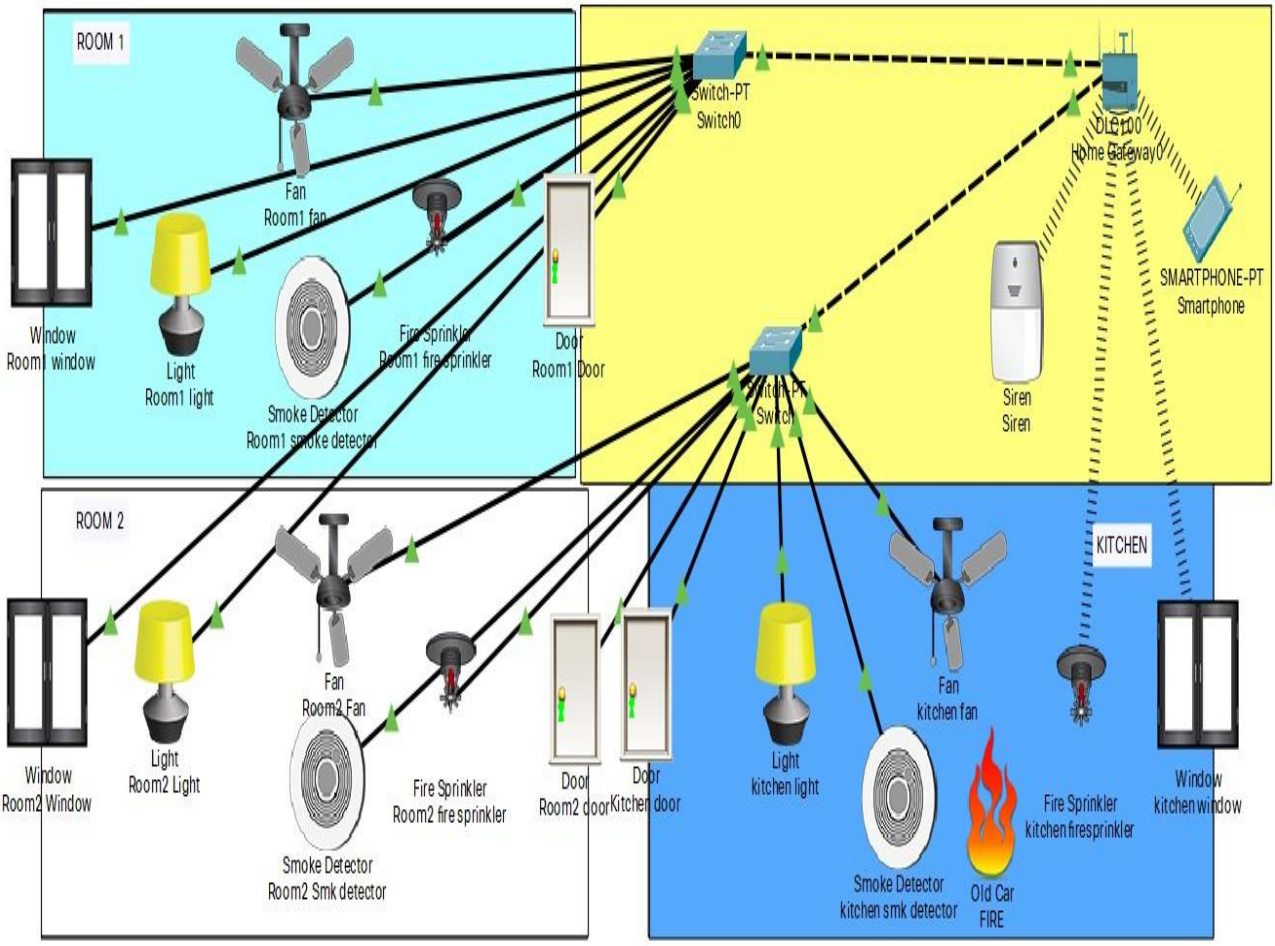


Figure 3.1 Fire Prevention System for Apartment with Packet Tracer for Cisco

The home gateway is used to assign IP addresses to smart devices and register them. The association with the cloud WAN is set up when the home door automatically receives an IP address from the ISP. Additionally, all the innovative hardware connected to the home doorway receives a natural IP address from the remote server. The home gateway is connected to the remote server using straight-through wires. The home gateway provides Java content, Python, and Visual Basic devices with a variety of programming environments. (Flifel 2019).

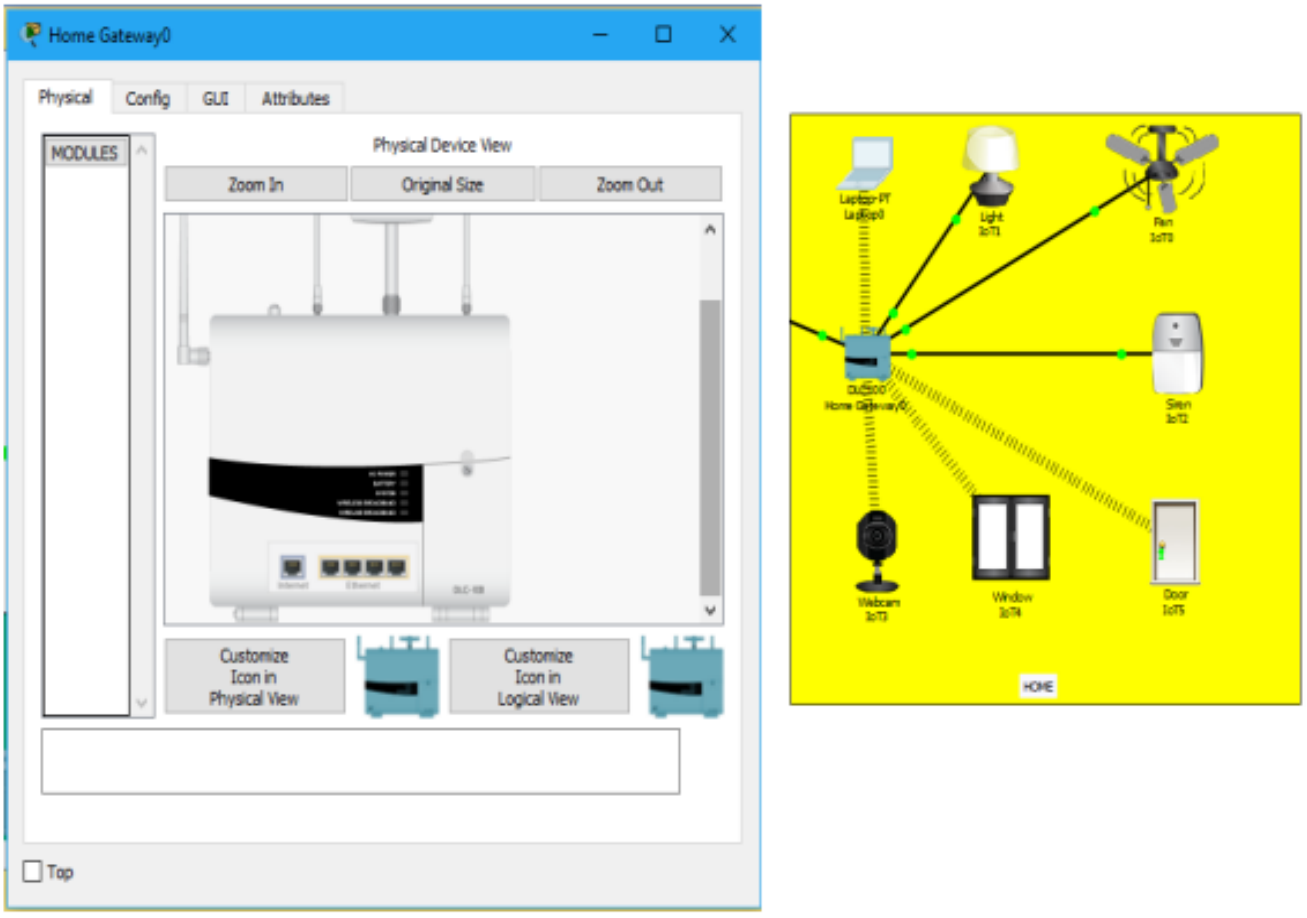


Figure 3.2 Home Gateway

Different smart objects connected to the home gateway using remote or Ethernet connection association can be shown in Figure 1.3 above. The front door of the house has four Ethernet connections and a remote door with the SSID "home gateway." The home gateway can be configured to use WEP/WPA-PSK/WPA2 protocols for the remote connection confirmation. Since the devices will be connected wirelessly, we should select "remote" when connecting them to the home gateway. After that, we identify the SSID of the home gateway on the devices.

The smart light setup for the home gateway is shown in Figure 3.3. To keep the setup simple, authentication has been turned off. We, therefore, repeated this arrangement on each of the devices.

5.3.1 Smart Light

The smart light arrangement for the home gateway is shown in Figure 3.3. To keep the settings simple, authentication has been turned off. Therefore, we used the same configuration on all of the devices.

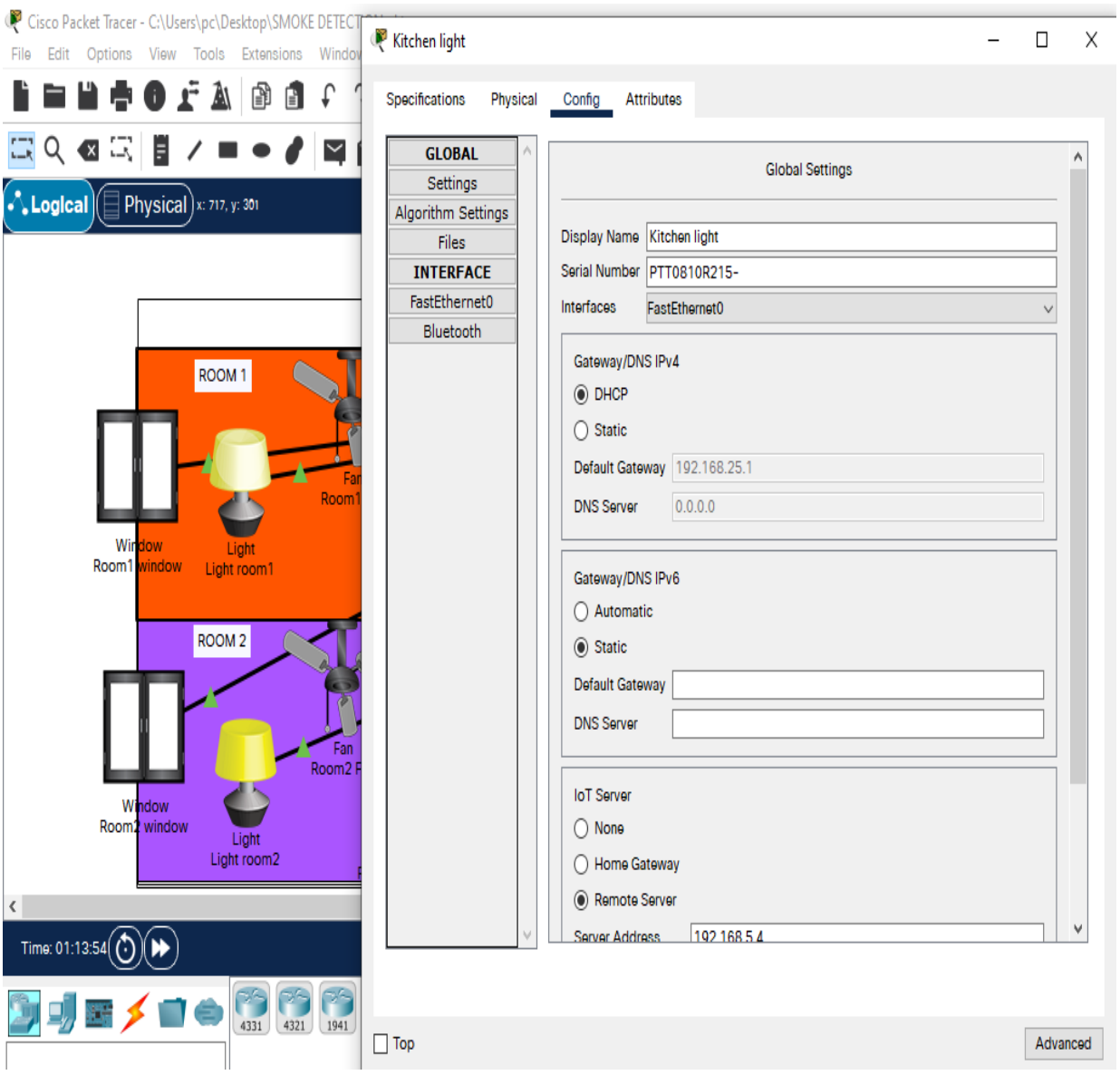


Figure 3.3 Connected to the Home Gateway is a Smart Light

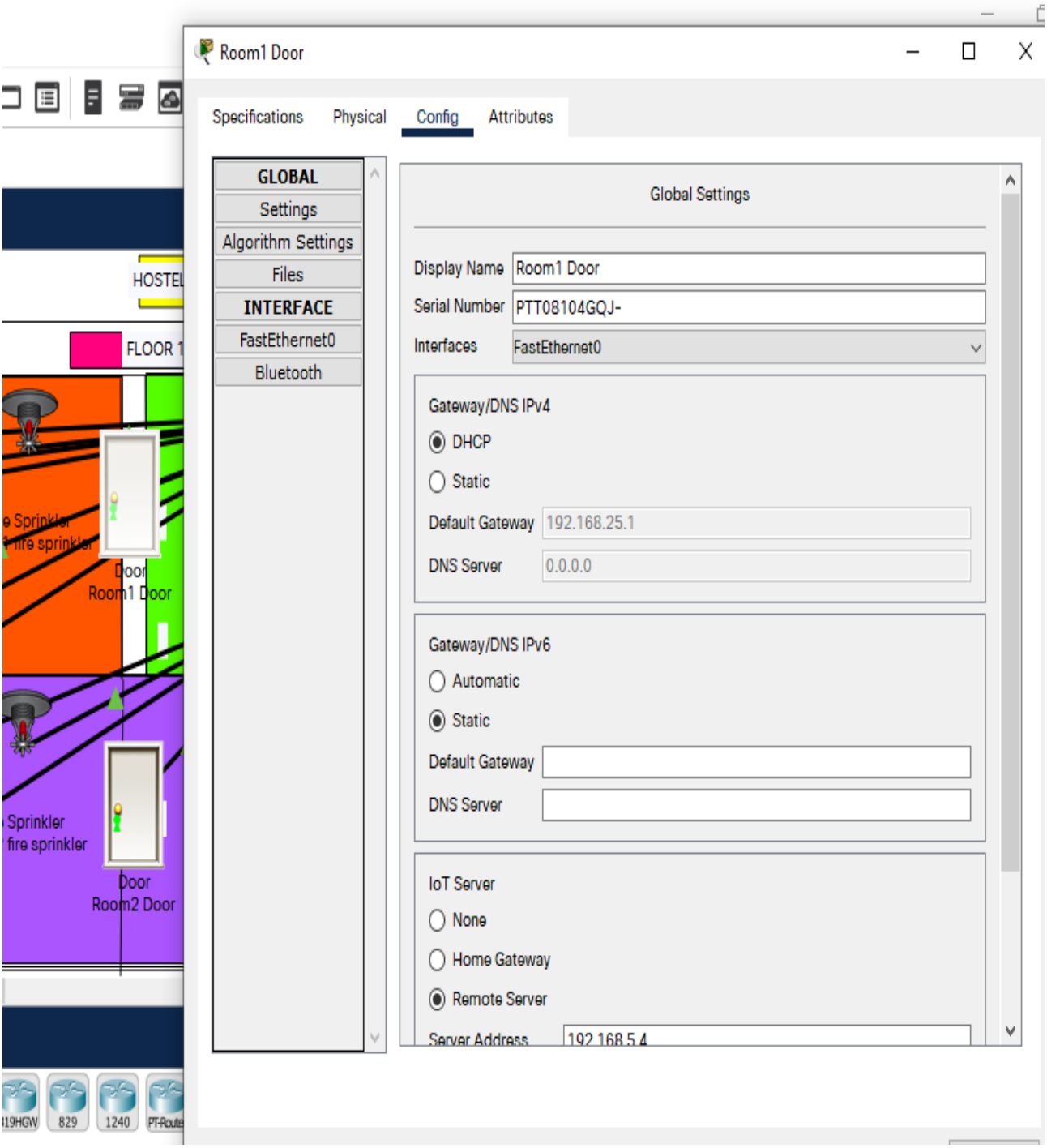


Figure 3.4 Room 1 IOT Door Configuration

5.3.2 Smartphone

Using a smartphone and a web connection, one can remotely access a smart product using a web interface at www.iot.org. (Khvoynitskaya 2020). To obtain an internet connection, the smartphone is linked to a cell tower. The appropriate APN (Access Point Name), "Purcell," is configured in the cell phone as shown in Figure 3.5 to connect the phone to the 3G mobile tower.

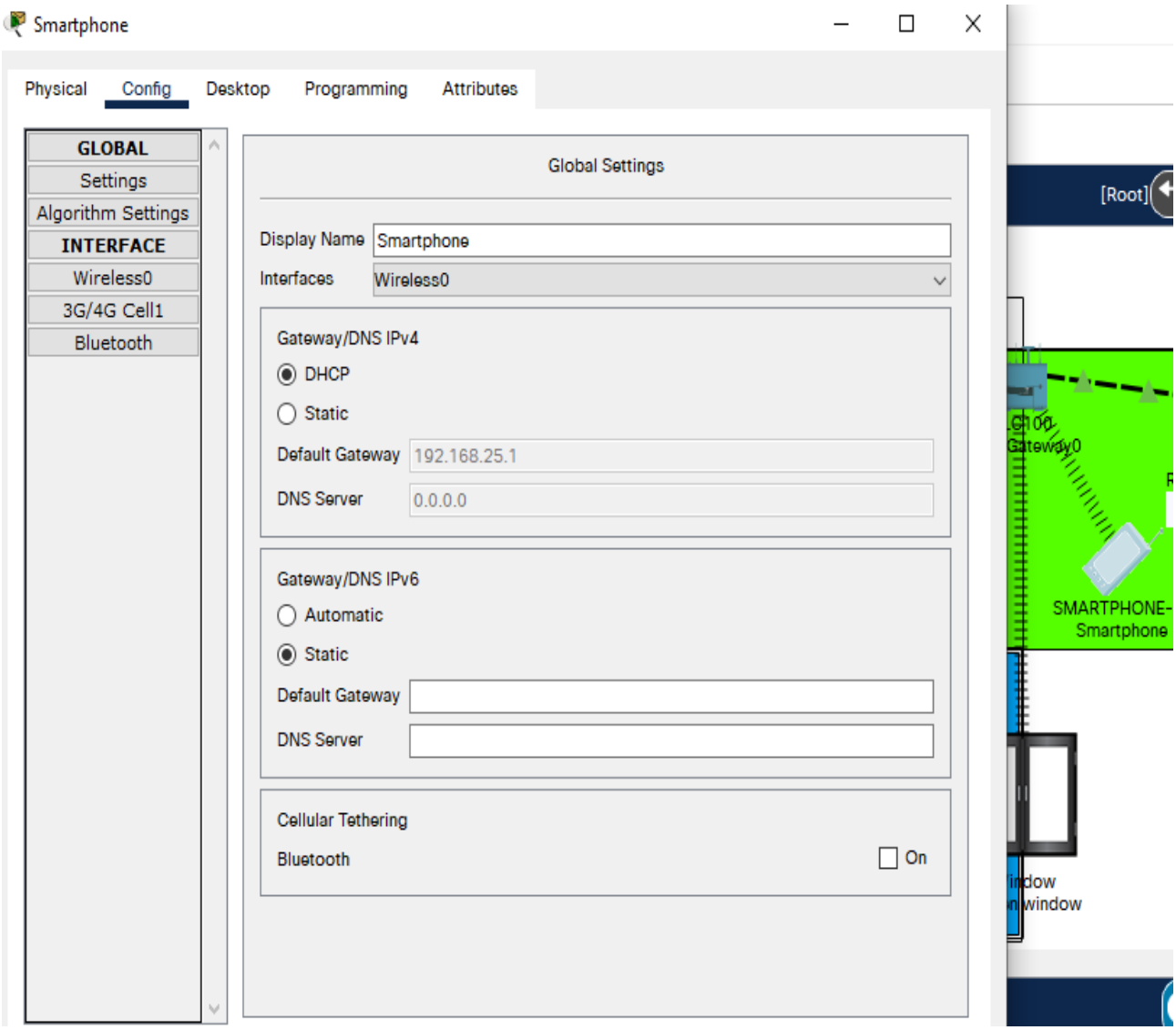


Figure 3.5 Smartphone Configuration View

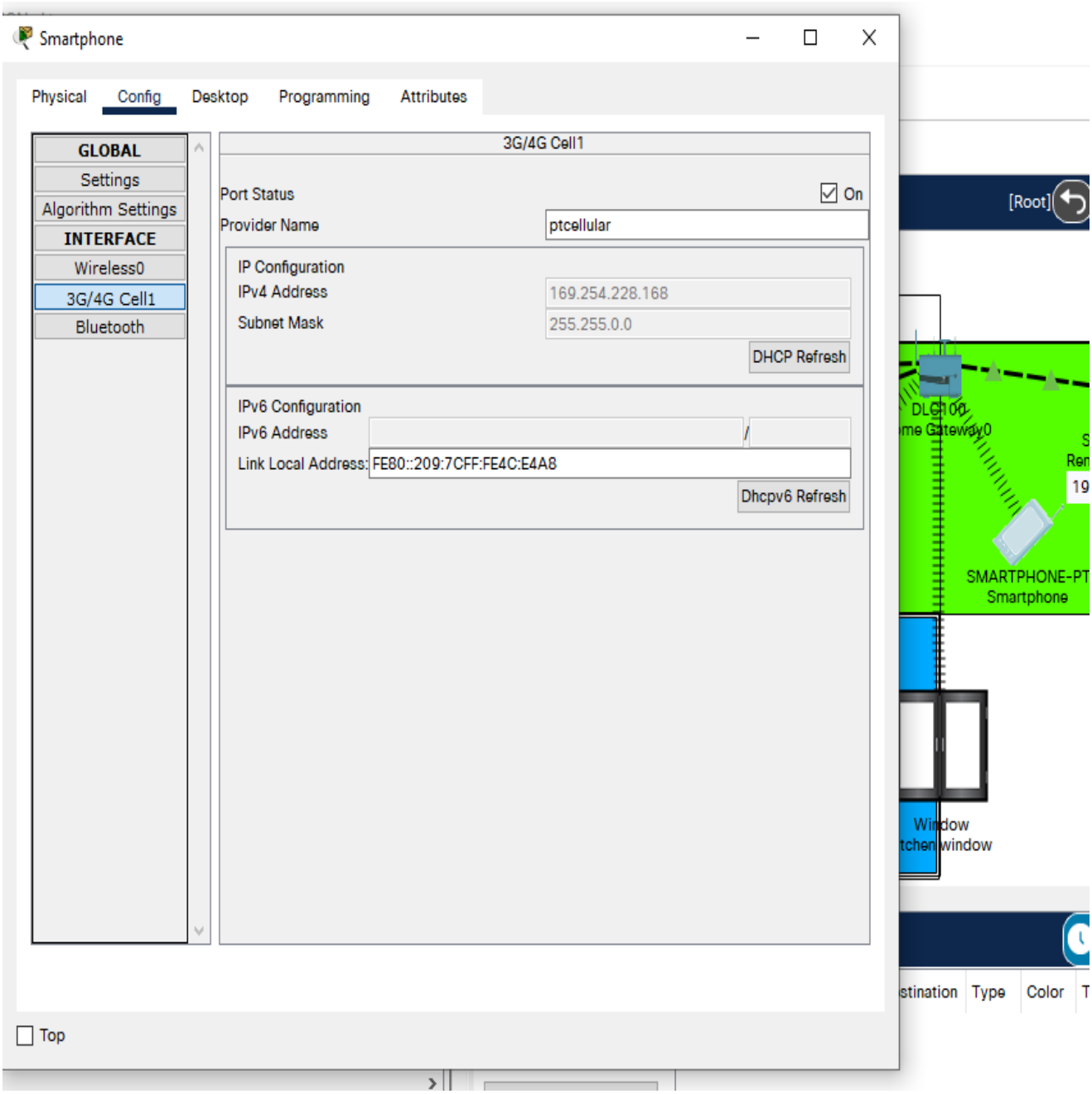


Figure 3.6 Smart Phone Connection to the Home Gateway

5.4 Result of implementation:

All of the devices can be accessed using the correct client's name and secret word on a smartphone's internet browser by entering the URL 192.168.6.1 after they have been configured.

The client can view a list of the related devices and perform the necessary action after connecting to the IoT home page through a browser and completing the authentication. Figure 3.6 depicts the registration home gateway login process using a smartphone's internet browser and the URL 192.168.5.4. The list of devices registered on the home gateway is shown in Figure 3.7

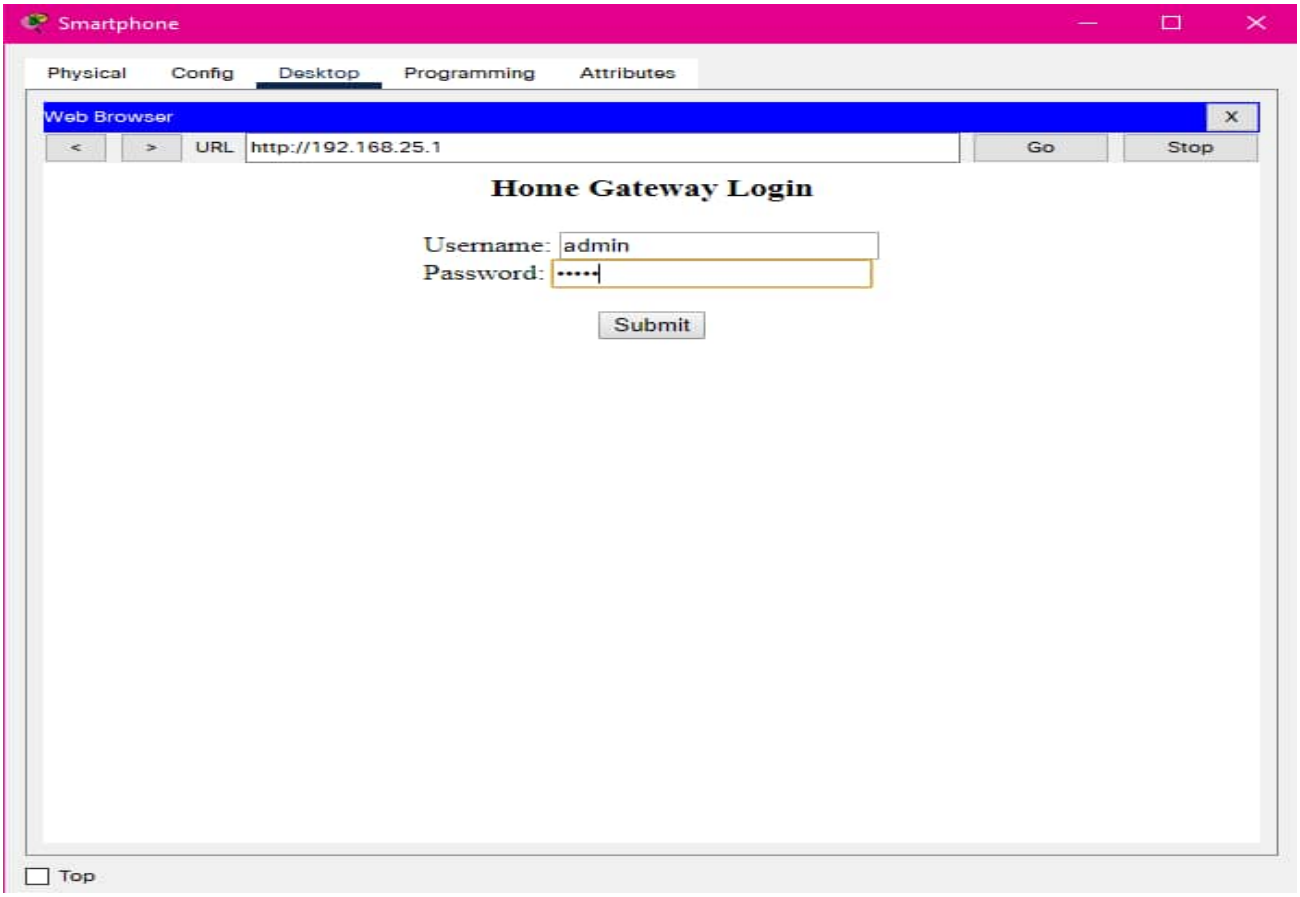


Figure 3.7 Home Gateway login

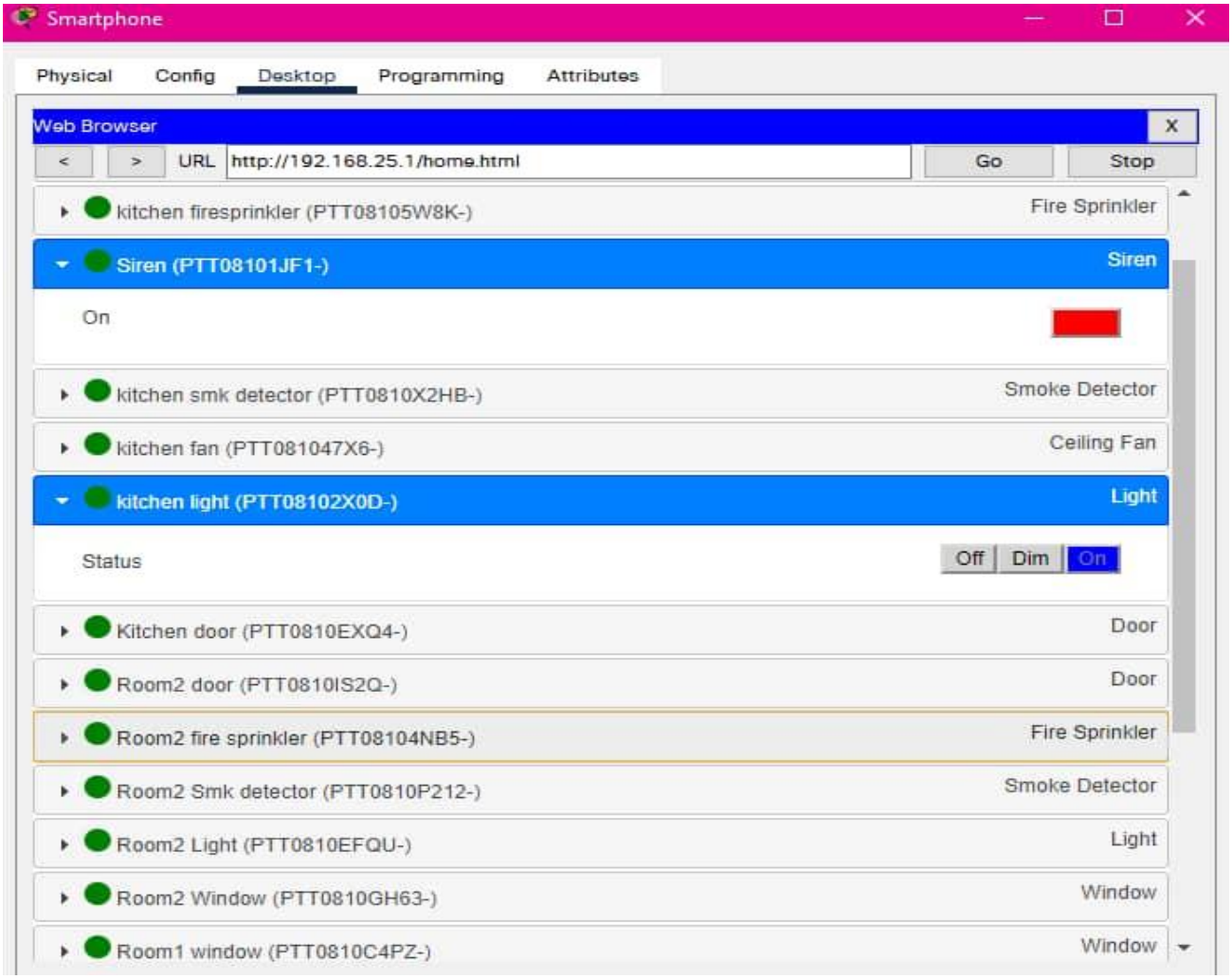


Figure 3.8 IOT Devices Status

5.6 Chapter Conclusion

The idea was to execute and reproduce a notable Internet of Things system that is fire prevention for apartments, utilizing packet tracers from Cisco. The execution is carried out with the most recent version of Cisco Packet Tracer (Cisco Packet Tracer 8.0), considering the way this version includes many smart gadgets utilized by the system.

CHAPTER 5

SUMMARY, CONCLUSION, AND RECOMMENDATION

6.1 SUMMARY

The fire prevention system aims to improve convenience, safety, and cost reduction for apartment developers and the real estate industry as a whole. It is a computer-based system that can help apartment developers to increase life safety, reduce fire outbreaks, and improve property security. The fire prevention systems are designed to ensure the security of property and the preservation of human lives. The data collected during the information-gathering process is properly analyzed, so the result is a prerequisite for the new system. The system has been tested and found to function normally, so the results produced by this technology are encouraging. Therefore, compared with the current system, the system will reduce life loss through fire outbreaks. It will also reduce property loss during fire outbreaks. The system will detect the onset of fire quickly and act accordingly to curb the potential fire outbreak.

6.2 CONCLUSION

The goal of this research was to use a Cisco packet tracer to replicate internet activities. Another modern and cutting-edge invention is the internet of things. Therefore, it was important to create a virtual tool where students could learn about and understand this technology. That is the reason for conducting this evaluation. I chose the Cisco packet tracer because it provides a simulation environment with devices that resemble actual devices. In addition, the new version of the packet tracer includes various Internet of Things devices, actuators, and various sensors, making it a useful test framework for the Internet of Things. The idea was to create and replicate the fire protection system, which is a very well-known Internet of Things application. The most recent version of the Cisco Packet Tracer (version 8.0) was chosen for the execution because it includes several intelligent devices for fire prevention. The reproduction is accomplished using a variety of different network equipment, including a gateway, switch, and smartphone.

The home gateway is used to connect various smart devices to it and distribute IP addresses to those devices through a remote network. The Home gateway and the mobile device are very important in the simulation since they allow for remote internet-based control of the IoT devices. The house door is used to register smart devices, and the smartphone is used to remotely access the devices registered on the home gateway. The inclusion of multiple internets of things and networking tools made the simulation easier, and more IoT tools will be added for the upcoming version of the Cisco packet tracer, allowing for the creation of more complex IoT simulations.

6.3 RECOMMENDATION

A lot of work and effort was put into the structuring and designing of this system (Fire Prevention System). It began with the gathering of ideas and information and progressed through many further steps until completion. Based on the benefits of this system and the tremendous value it will provide, the following recommendation will be considered: IoT technology is simulated using a variety of simulators. Therefore, future studies might compare Cisco Packet Tracer to other Internet of Thing's simulators like NS3 or GNS3, among others. Any research into other IoT simulators was done as part of this project because it was only focused on using Cisco Packet Tracer to simulate IoT. Since it is anticipated that Cisco will release a new version of packet tracing with more IoT devices, future studies can simulate more complicated IoT applications.

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APPENDIX

USER GUIDE

For effective use of this system, the user must have basic computing knowledge.

To install the Cisco packet tracer software used in building the system, a minimum of Windows 7 is required.

The user must meet the following hardware requirements:

Hard disk space of 500 Megabytes (MB)

1 gigabyte (GB) of RAM.

CPU: Intel Pentium 4